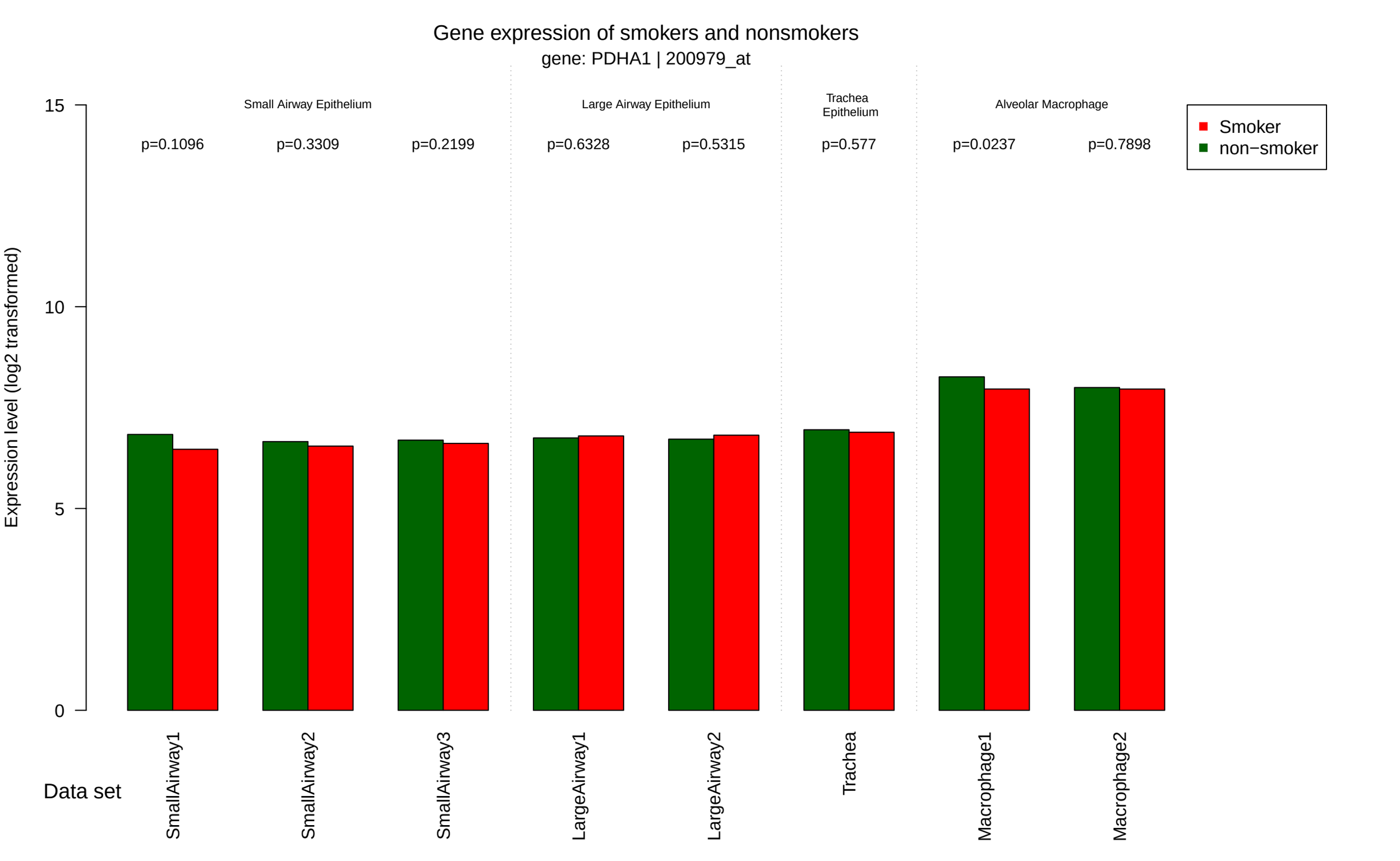
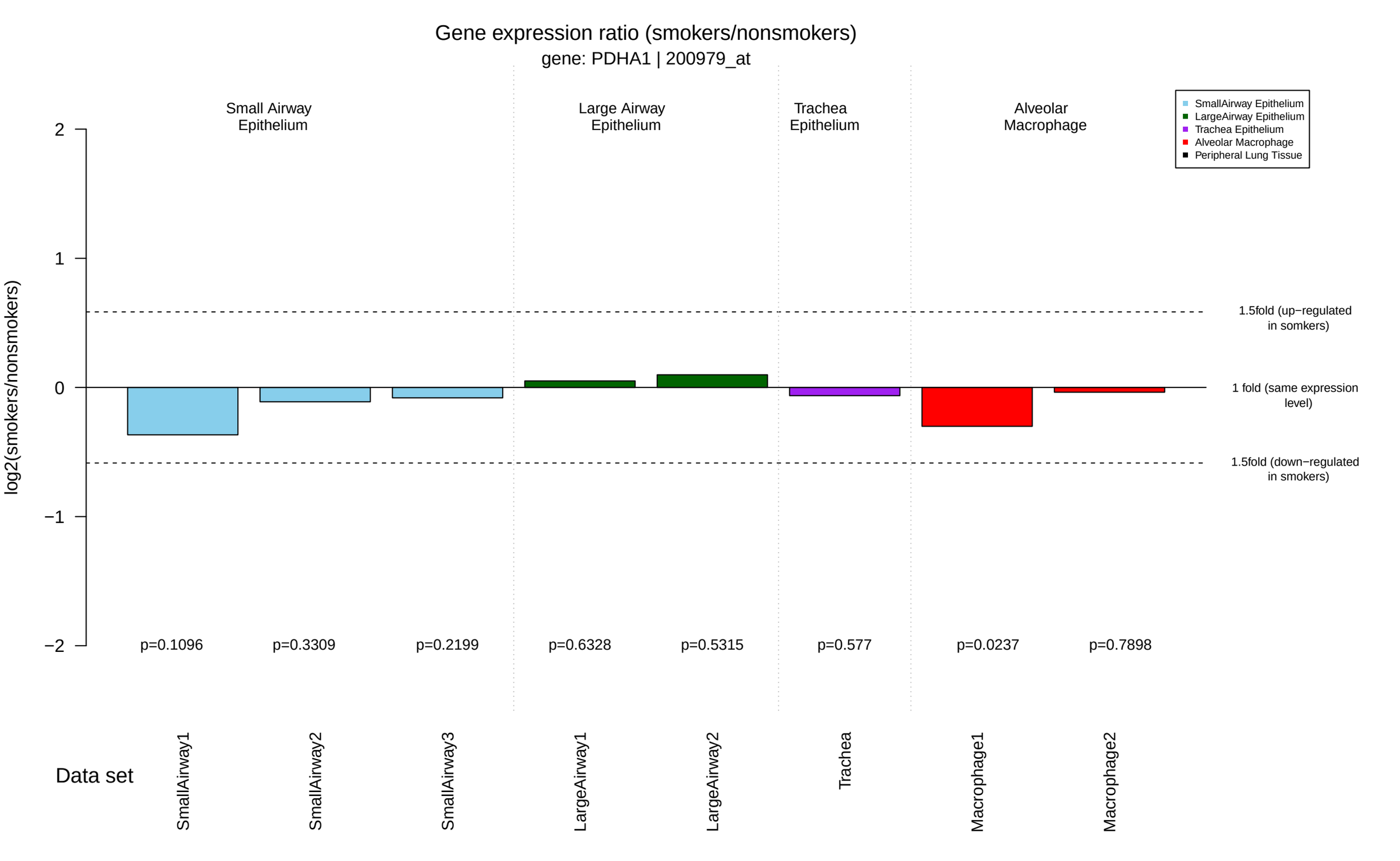
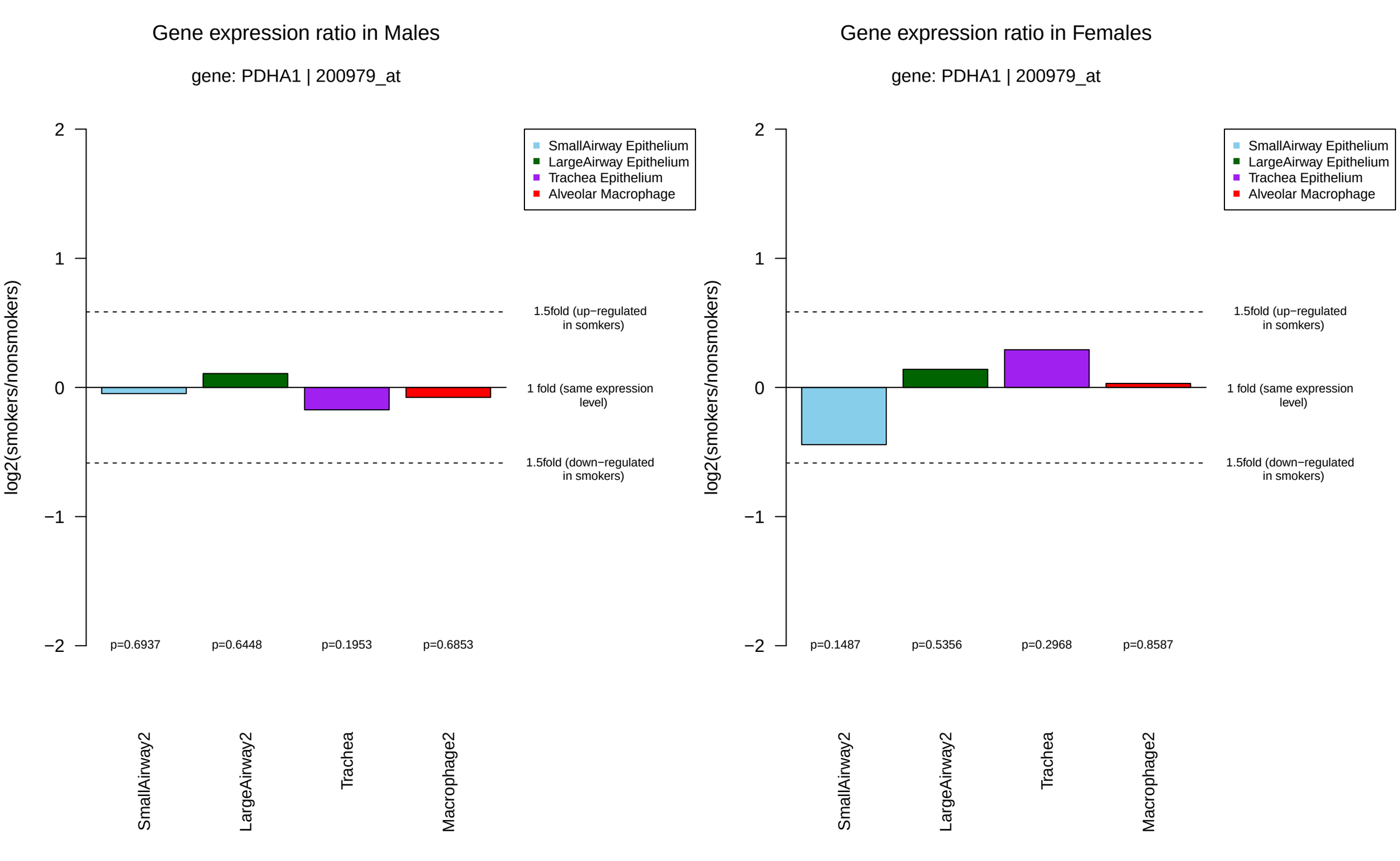
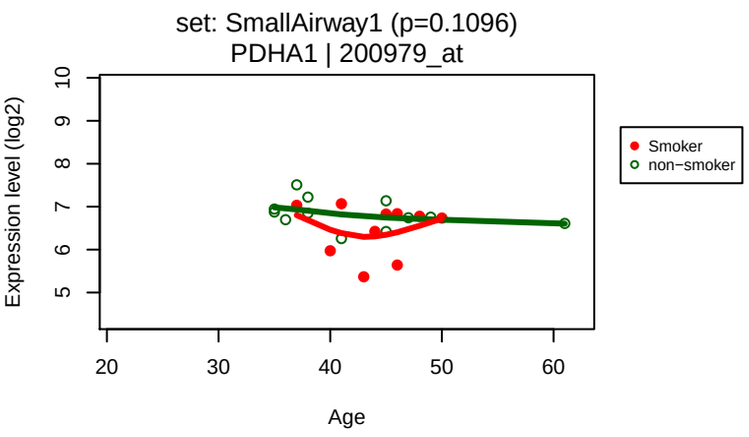
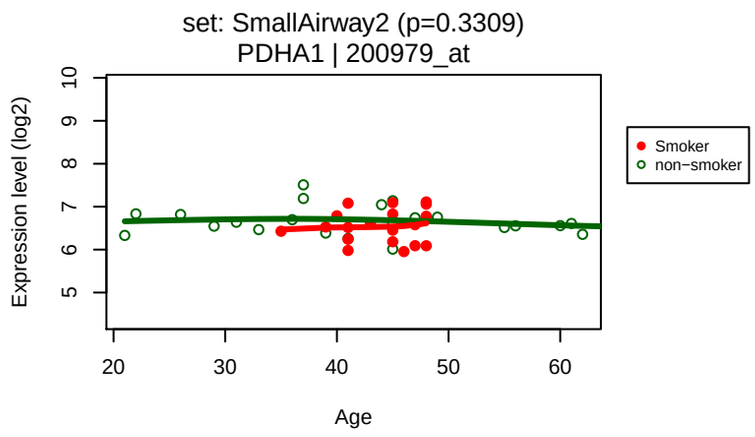
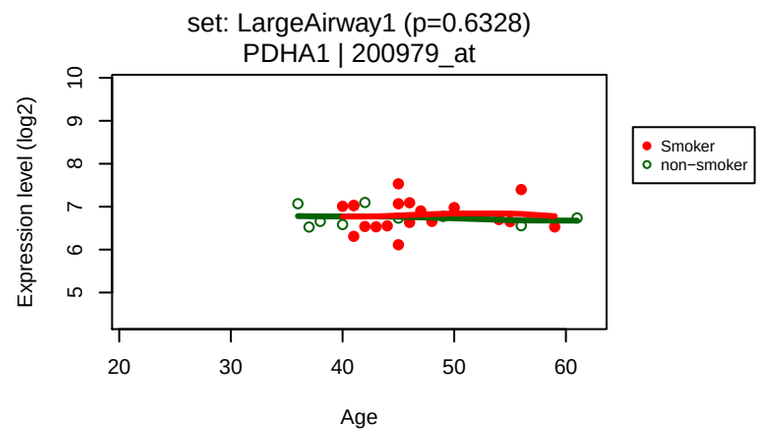

Gene expression of smokers and nonsmokers
gene: PDHA1 | 200979_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1096
p=0.3309
p=0.2199
p=0.6328
p=0.5315
p=0.577
p=0.0237
p=0.7898
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

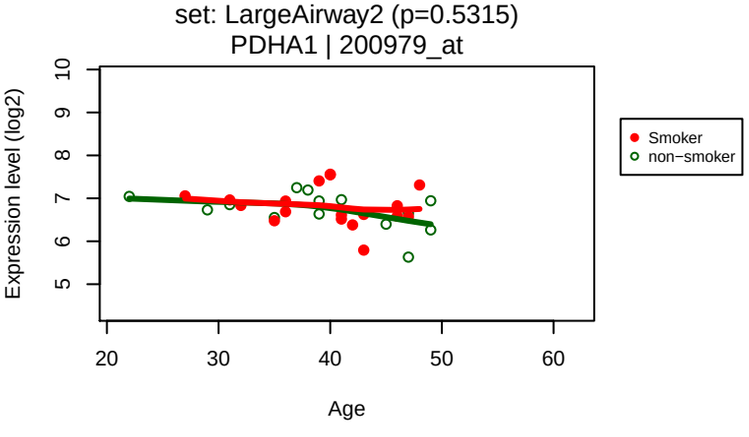

Gene expression ratio (smokers/nonsmokers)
gene: PDHA1 | 200979_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1096
p=0.3309
p=0.2199
p=0.6328
p=0.5315
p=0.577
p=0.0237
p=0.7898
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

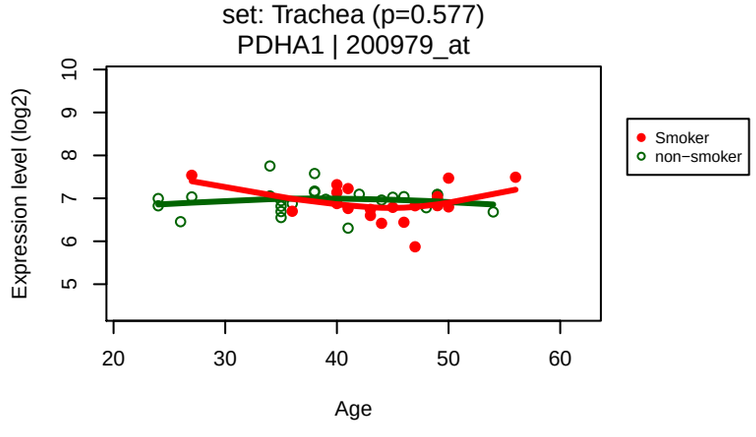

Gene expression ratio in Males
gene: PDHA1 | 200979_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6937
p=0.6448
p=0.1953
p=0.6853
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDHA1 | 200979_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1487
p=0.5356
p=0.2968
p=0.8587
SmallAirway2
LargeAirway2
Trachea
Macrophage2

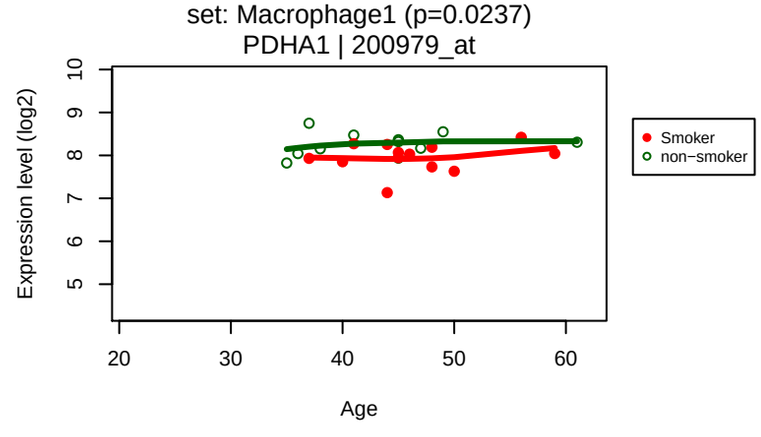

set: SmallAirway1 (p=0.1096)
PDHA1 | 200979_at
Expression level (log2)
Age
Smoker
non−smoker

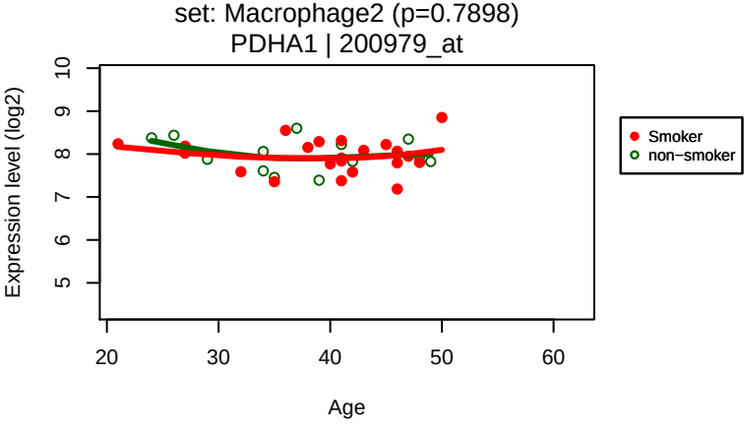

set: SmallAirway2 (p=0.3309)
PDHA1 | 200979_at
Expression level (log2)
Age
Smoker
non−smoker

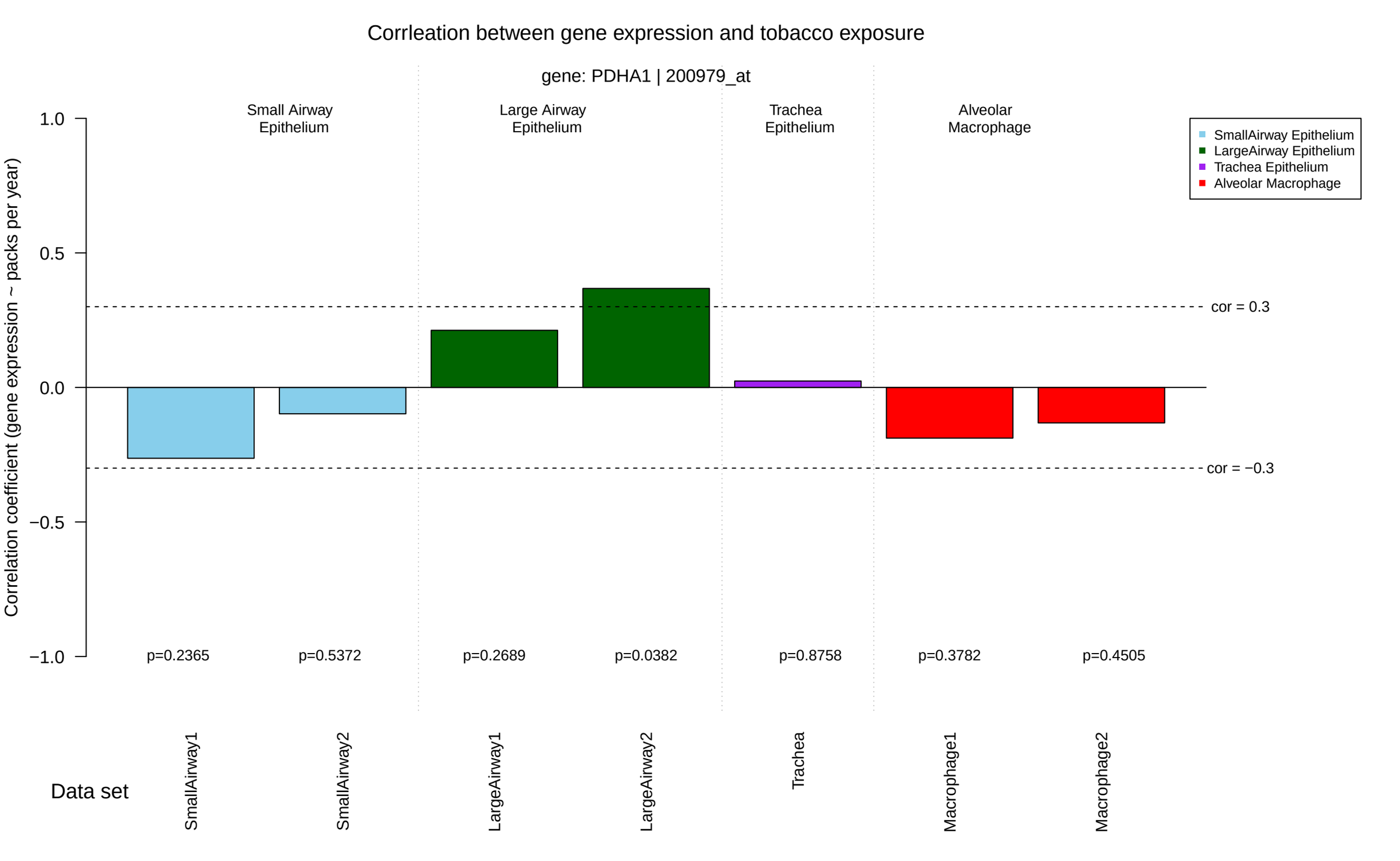

set: LargeAirway1 (p=0.6328)
PDHA1 | 200979_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5315)
PDHA1 | 200979_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.577)
PDHA1 | 200979_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0237)
PDHA1 | 200979_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7898)
PDHA1 | 200979_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDHA1 | 200979_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2365
p=0.5372
p=0.2689
p=0.0382
p=0.8758
p=0.3782
p=0.4505
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2